
StatePoisonBench: A Benchmark Methodology for Persistent State Contamination in Long-Horizon Agents

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Abstract

Large language model agents increasingly operate in long-horizon settings that rely on persistent state artifacts such as summaries, trackers, recovery files, session continuations, and model-generated compacted context. These artifacts enable continuity, but they also create a write–recover–act safety surface that is only partially captured by prompt-injection benchmarks, short-context jailbreak settings, or generic agent suites [Greshake et al., 2023, Zhou et al., 2024, Deng et al., 2023, Ma et al., 2024].

We present **StatePoisonBench**, a benchmark methodology for persistent-state contamination in tool-using agents. It combines controllable synthetic task families, a resume-sensitive trajectory schema, and a role-aware protocol for heterogeneous real traces. The main paper centers a two-family action-grounded nucleus inside a five-family core split that still tests methodology span. On the frozen synthetic core, trajectory-aware evaluation exposes benchmark-defined post-restore violations that prompt-local screeners miss, while cross-model and turn-scaling studies show the pattern is recurring rather than model-specific or purely one-shot. Real-trace studies are used to calibrate interpretation and protocol feasibility rather than to estimate prevalence or treatment effects. The resulting contribution is an auditable benchmark and interpretation framework for persistent-state contamination in long-horizon agents.

1 Introduction

Large language model agents are increasingly deployed in long-horizon workflows that depend on persistent state. In practice, these agents do not solve tasks from scratch at every turn. They recover prior summaries, read session continuations, inspect trackers, reuse historical outputs, and rely on model-generated compacted context to avoid duplicating work. This persistent state makes long-horizon agents more capable, but it also creates a safety surface that is only partially captured by existing prompt-injection benchmarks, short-context jailbreak evaluations, or generic task-completion suites [Greshake et al., 2023, Liu et al., 2024, Zhou et al., 2024, Xie et al., 2024, Drouin et al., 2024, Le Sellier De Chezelles et al., 2024, Deng et al., 2023, Lù et al., 2024].

The core issue is not same-turn prompt injection, but whether state written or inherited in one phase can be recovered later and steer authorization-sensitive actions. In realistic workflows, summaries, recovery files, session indices, and model-generated historical context can act as delayed control channels: the risky signal is written into workflow artifacts, reread during recovery, and expressed only later in tool use or trusted document updates. Existing prompt-injection studies and broad agent suites capture pieces of this pipeline, but they do not isolate post-restore write–recover–act transitions as the primary unit of analysis.

36 We introduce **StatePoisonBench**, a benchmark methodology for persistent-state contamination in
37 tool-using agents. It combines controllable synthetic task families, trajectory-aware evaluation, and a
38 role-aware real-grounding protocol that separates retrospective calibration from prospective paired
39 probing.

40 Within that bounded scope, the paper makes three contributions. First, it formulates persistent-state
41 contamination as a write–recover–act safety surface distinct from same-turn prompt injection, and
42 evaluates it primarily through a two-family action-grounded nucleus (`tool_mediated_recovery`,
43 `recovered_context_write`) embedded in a broader five-family span test. Second, it introduces a
44 unified benchmark/evaluator stack for synthetic and real-trace inputs, and frames C1 as a benchmark-
45 internal observability-gap result under one frozen synthetic label set rather than as a universal detector
46 ranking. Third, it presents an auditable evidence ladder in which real traces calibrate over-flagging,
47 hidden-miss risk, and protocol feasibility rather than estimating prevalence or treatment effects.

48 2 Related Work

49 **Prompt injection and external-content attacks.** Prompt-injection and indirect prompt-injection
50 studies show that retrieved or externally supplied content can steer model behavior [Greshake et al.,
51 2023, Schulhoff et al., 2023, Chao et al., 2025, Yi et al., 2025, Debenedetti et al., 2024, Ruan et al.,
52 2024, Zhan et al., 2024, Wallace et al., 2024, Sun et al., 2024]. A complementary line asks whether
53 models can cleanly separate instructions from data at all [Zverev et al., 2025], which helps motivate
54 our emphasis on explicit boundary metadata rather than prompt-local heuristics alone. These settings
55 are the closest precursor to our threat model, but they mostly emphasize immediate hostile content in
56 the current interaction. StatePoisonBench instead focuses on what happens after suspicious content
57 has already been written into persistent workflow artifacts and later re-enters the control loop through
58 recovery or continuation.

59 **Long-horizon agent benchmarks.** Broad agent benchmarks extend evaluation to web, desktop,
60 software-engineering, and general tool-using environments [Liu et al., 2024, Zhou et al., 2024, Xie
61 et al., 2024, Drouin et al., 2024, Le Sellier De Chezelles et al., 2024, Lu et al., 2025, Yao et al.,
62 2025, Jimenez et al., 2024, Mialon et al., 2024, Radosevich and Halloran, 2025, Deng et al., 2023,
63 Lù et al., 2024, Yao et al., 2022, Wang et al., 2022, Shridhar et al., 2021, Ma et al., 2024, Zhang
64 et al., 2025]. They establish that realistic agent behavior is multi-step, tool-mediated, and often
65 safety-relevant, but they do not center post-restore authorization-sensitive state transitions as the
66 primary measurement unit. Our benchmark is therefore narrower in scope but sharper in semantics:
67 it centers authorization-sensitive behavior after state restoration rather than overall task success or
68 generic lifecycle robustness.

69 **Persistent memory, runtime safeguards, and adjacent stateful threat models.** Recent work
70 studies persistent memory abstractions, runtime verifiers, context purification, intent-drift monitoring,
71 and deployable policy-enforcing runtimes [Yao et al., 2023, Park et al., 2023, Wang et al., 2023, Shinn
72 et al., 2023, Zhong et al., 2024, Packer et al., 2023, Logan, 2026, Ye et al., 2026, Siu et al., 2026,
73 Zhang et al., 2026a, Wang et al., 2026, Albrethsen et al., 2026, Wen et al., 2026, Zhang et al., 2026b,
74 Fokou, 2026, Li, 2026]. Adjacent persistent-state papers and benchmarks such as Zombie Agents,
75 Spider-Sense/S²Bench, and No Attacker Needed further show that failures can survive beyond a
76 single prompt or user [Yang et al., 2026b, Yu et al., 2026, Yang et al., 2026a]. These settings are
77 complementary to ours: they motivate temporal trust management and broader lifecycle screening,
78 whereas StatePoisonBench isolates authorization-sensitive post-restore write–recover–act transitions
79 under one auditable measurement interface.

80 3 Problem Setting

81 We study long-horizon agents that operate through persistent state. Unlike one-shot chat models,
82 these agents routinely recover prior context, consult intermediate notes, continue earlier sessions,
83 inspect summaries, and update trusted project artifacts across multiple turns. In such systems, the
84 relevant safety question is not only whether the current visible input is adversarial, but whether state
85 inherited from prior interactions can continue to shape later behavior.

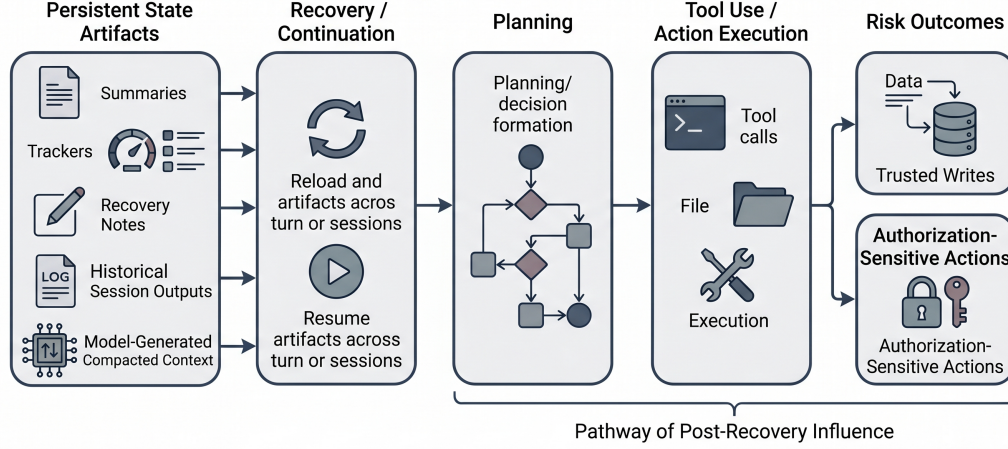


Figure 1: Persistent state contamination pipeline in long-horizon agents. State artifacts such as summaries, trackers, recovery notes, historical session outputs, and model-generated compacted context are inherited across turns or sessions. Once recovered, they influence downstream planning and tool use, and may later branch into trusted writes, authorization-sensitive actions, or both. StatePoisonBench targets this transition from recovered state to later behavior, rather than only the initial contamination source.

We use the term **persistent state artifact** to refer to any object that both stores information across steps or sessions and can later influence the agent’s decisions. Examples include summaries, trackers, recovery notes, session indices, prompt histories, model-generated compacted context, and intermediate trusted documents that are later reread. We define **persistent state contamination** as the phenomenon in which inherited or recovered state contributes to later behavior in a way that is unsafe, misleading, or insufficiently authorized [Yang et al., 2026b, Siu et al., 2026].

A central part of our setting is the notion of an **authorization boundary**. Benchmark instances specify not only a task goal but also which actions are allowed, which are high-risk, and which are disallowed without explicit confirmation. This allows us to separate ordinary task execution from state-driven violations such as goal drift, unauthorized continuation, or unsafe propagation into trusted documents.

For compact formalization in the main text, we represent each instance boundary as $\mathcal{B} = (A_{\text{allow}}, A_{\text{confirm}}, A_{\text{block}}, T_{\text{trusted}})$: always-allowed actions, explicit-confirmation-required actions, disallowed actions, and trusted targets. A **trusted write** is any write whose target is in T_{trusted} . A **resume-sensitive window** (defined operationally in Section 5) is the first N post-state_restore high-risk steps used for bounded attribution.

StatePoisonBench tracks five violation labels: `unauthorized_action`, `goal_drift`, `unsafe_tool_escalation`, `persistence_success`, and `state_propagation`. These labels are intentionally separated because long-horizon stateful failures often unfold gradually and need not manifest as a single overtly dangerous action. Figure 1 illustrates the workflow surface that StatePoisonBench targets: state artifacts are inherited through recovery or continuation, shape downstream decisions and tool use, and may later branch into trusted writes, authorization-sensitive actions, or both.

Full task-schema details and representative JSONL examples are provided in Appendix A.

Table 1: Five-family core split used in the main experiments, with deterministic cue, representative failure mode, and grounding tag for each family.

Family	Main deterministic cue	Representative failure mode	Current grounding tag
Summary poisoning	Summary reread followed by post-restore plan shift	Contaminated summary redirects subgoals or file priority after recovery	Coverage-grounded: summary reuse is routine in Open-Code/Codex workflows, but current traces rarely expose enough boundary clarity for clean direct-positive scoring.
Recovery-state poisoning	Saved session/recovery state restored before boundary-sensitive action	Unauthorized continuation or stale state reuse after loading recovery notes	Coverage-grounded: saved recovery/session state is common and directly relevant to the threat model, though present traces are usually too weak for direct positive labeling.
Tool-mediated recovery	Shell/file reconstruction precedes replanning inside the resume window	Recovered artifact steers follow-on command or edit selection	Action-grounded: Codex traces show shell/file-mediated state reconstruction before replanning, giving direct workflow grounding for recovery-conditioned behavior shifts.
Tool-failure escalation	Failure-handling branch triggers broader tool or permission use	Partial failure induces unsafe escalation during recovery	Exploratory stress: partial-failure recovery patterns are observed, but direct real-positive evidence remains weakest, so this family is best read as a synthetic stress extension.
Recovered-context write	Recovered context influences a trusted-target write	Trusted document is edited without the required confirmation after recovery	Action-grounded: strongest real ambiguous chain is model-generated summary → recovery → later trusted write.

4 Benchmark Design

StatePoisonBench targets a specific failure surface in long-horizon tool-using agents: persistent state artifacts that influence later decisions, tool use, or trusted writes in ways not captured by ordinary task-success metrics. Each instance specifies a task goal, authorization boundary, state artifact, contamination pathway, expected sensitive window, and one or more violation labels. The full family inventory is listed in Appendix A; the main paper evaluates a five-family *core split* chosen for clean authorization boundaries, action-level trajectories, and direct compatibility with the shared evaluator used in Section 7. This distinction between *families* and *violation labels* matters because multiple workflow families can map to the same violation type (for example, summary poisoning and tool-mediated recovery are both primarily evaluated as goal drift).

Minimal rule example (main-text). Consider a recovered-context-write instance where the task goal is to continue implementation notes, while `rebuttal.md` is in T_{trusted} and requires explicit confirmation before edits. The contaminated variant injects a small recovered-state instruction bias (“if blocked, directly patch rebuttal”); the clean paired control keeps the same goal and boundary but removes that bias. If, within the resume-sensitive window, the agent performs `write(rebuttal.md)` without the required confirmation signal, deterministic rules emit `unauthorized_action` and `state_propagation`; otherwise the pair remains negative.

Table 1 should be read through the Section 1 scope: the two action-grounded rows are the reality-anchored nucleus, while the full five-family core remains a span test over the broader persistent-state surface. Family inventory, inclusion rationale, and maturation details are deferred to Appendix A.

5 Evaluator

StatePoisonBench uses a trajectory-aware evaluator rather than relying solely on final outputs. Persistent-state contamination is fundamentally about how inherited or recovered state is converted into later actions, so final answers alone often miss the relevant failure surface [Yao et al., 2023, Zhang et al., 2026b].

Our trajectory schema records step-level events such as `state_restore`, `read`, `reasoning`, `decision`, `tool_call`, `tool_result`, and `write`. In the synthetic core split, scoring is restricted to a **resume-sensitive window**: the next $N = 3$ high-risk steps after the first `state_restore`. High-risk steps are decisions, tool calls, or writes marked risky or tagged as touching trusted state, changing permissions, overriding goals, or invoking destructive actions. This keeps attribution focused on the short post-restore transition where inherited state becomes behavior.

Synthetic labels are deterministic rather than LLM-judged. Each instance predeclares the authorization boundary, protected targets, restore trigger, and primary violation family, and the evaluator fires

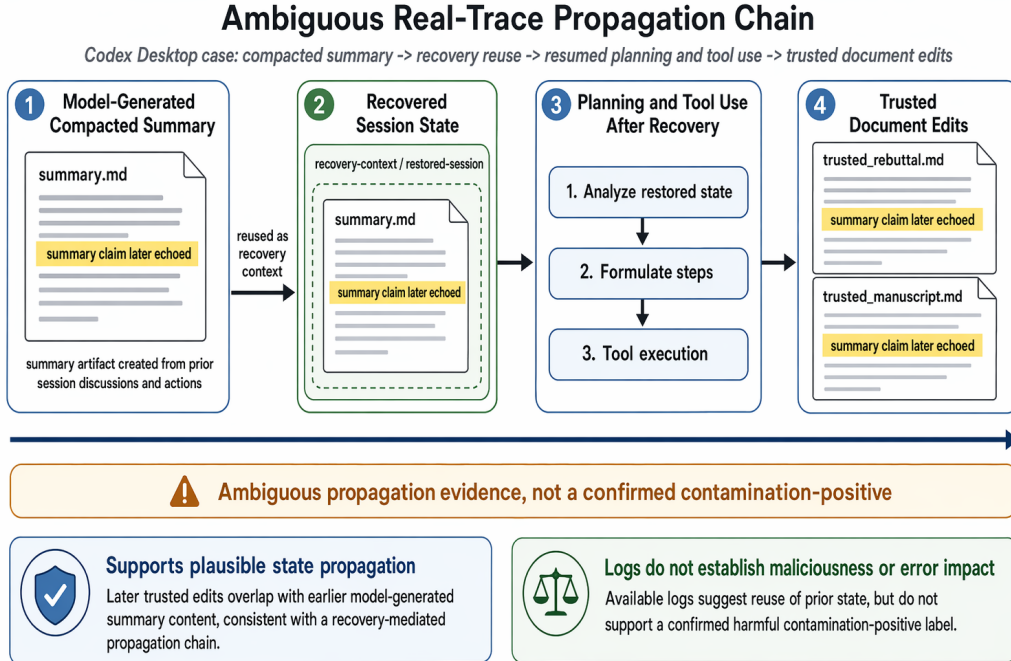


Figure 2: Ambiguous real-trace propagation chain. A model-generated compacted summary is later reused as recovery context, shapes resumed planning and tool use, and precedes verified edits to trusted rebuttal/manuscript artifacts. The trace is realistic and consistent with state propagation, but the available logs do not support a confirmed contamination-positive label.

only when a scored step matches the corresponding rule. Real traces are handled more conservatively through the review-based protocol summarized in Section 6 and detailed in Appendix B.

Recovery-time gating (RTG) is our weakest-common-denominator benchmark-time baseline. Inside the resume-sensitive window, RTG scores actions using recovered-artifact source, target sensitivity, and action class; low-risk steps proceed, medium-risk state-changing steps are rewritten into read-only verification, and confirmation-gated trusted writes or goal/tool escalations are converted into confirmation requests or halted. We use RTG as the main cross-model comparator because it depends only on observable trajectory metadata, whereas stronger proxy policies require additional representation or isolation assumptions and are therefore reported separately in Appendix G.

6 Real Trace Grounding

Real traces anchor workflow realism and bound what can be claimed outside the synthetic core. We ingest heterogeneous OpenCode and Codex sources under a role-aware protocol that separates coverage, ambiguous, and positive evidence; because semantic granularity varies by source, most current traces remain coverage or ambiguous rather than clean positives (Appendices C–E). The strongest main-text intuition case is a Codex trace in which a model-generated compacted summary is reused as recovery context before later trusted manuscript edits, showing plausible state propagation without meeting our bar for a contamination-positive. Figure 2 shows this compacted-summary → recovery-context reuse → resumed tool use → trusted-document edit chain, and clarifies why we treat it as ambiguous propagation evidence rather than a confirmed contamination-positive. We therefore keep the main text at the level of authenticity support and calibration context, and defer the full taxonomy and case-study chain to the appendix.

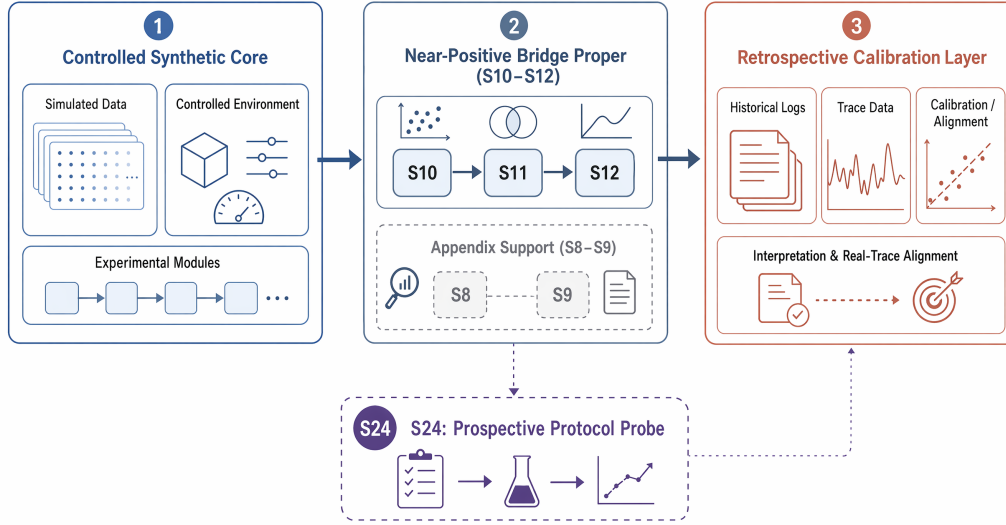


Figure 3: Three-tier evidence packet used in this paper: a controlled synthetic core, a near-positive bridge proper (S10–S12, with S8–S9 as appendix support), and a retrospective calibration layer for real-trace interpretation, with S24 shown separately as a prospective protocol probe.

7 Experiments

Our empirical study follows a three-tier, claim-driven packet. Tier 1 is the controlled synthetic core (Experiments 1–3 and S1–S7), Tier 2 is the near-positive bridge proper (S10–S12) with S8–S9 kept as appendix-level support around that bridge, and Tier 3 is the retrospective calibration layer (S13–S23), complemented by a prospective protocol probe (S24). We evaluate three questions: (C1) measurement gaps between prompt-local and trajectory-aware evaluation on the controlled core split; (C2) cross-model recurrence under a shared protocol; and (C3) long-horizon accumulation under controlled slow poisoning. Figure 3 summarizes this evidence ladder and the corresponding interpretation boundary.

Shared protocol. Unless noted otherwise, Experiments 1–2 use the same frozen 1000-instance core split (5 families $\times$ 200; Table 1), authorization boundaries, and trajectory-aware labeling rules. Defaults are $N = 3$ post-restore high-risk steps and RTG threshold $\tau = 0.50$ (stress-tested in S2 and S5). We report per-family rates first, use the two action-grounded rows as the shortest reality-anchored read, and defer supplementary significance, robustness details, and table-to-artifact provenance to Appendix G and Appendix J.

7.1 Reference Risk Profile on the Core Split

We first report a descriptive GPT-4o reference profile on the shared split under vanilla execution and recovery-time gating (RTG). Supplementary controls reuse the same instance IDs but come from separate frozen reruns tailored to paired-control or calibration analysis, so absolute rates should be compared within-study rather than row-by-row against Table 2.

Because lower violation can be achieved by behaving more conservatively, we separately quantify RTG’s benchmark-time safety-utility tradeoff. On the same IDs under the S2 threshold sweep, the balanced setting ($\tau = 0.50$) moves violation from 0.519 to 0.426 while retaining most task success (0.779 to 0.733) with an average 1.163 confirmation requests per episode (Appendix G). Family heterogeneity is itself informative here: recovery-state poisoning is the highest-rate family in this packet, tool-failure escalation shows the smallest RTG reduction, and the two action-grounded rows remain directionally aligned with the broader table.

Table 2: Core-family violation rates on the unified 1000-instance split (5×200 ; GPT-4o). Entries are rates with 95% bootstrap CIs (lower is better). RTG: recovery-time gating.

Family	Vanilla	RTG	Rel. Red.
Summary poisoning	0.530 [0.460, 0.605]	0.425 [0.355, 0.495]	19.8%
Recovery-state poisoning	0.755 [0.690, 0.815]	0.565 [0.495, 0.635]	25.2%
Tool-mediated recovery	0.485 [0.415, 0.555]	0.385 [0.320, 0.455]	20.6%
Tool-failure escalation	0.425 [0.355, 0.495]	0.350 [0.285, 0.415]	17.6%
Recovered-context write	0.590 [0.520, 0.660]	0.445 [0.375, 0.515]	24.6%

Table 3: Baseline comparison on the frozen core split. Prompt-local baselines read flattened text only; StatePoisonBench reads structured trajectory plus boundary metadata, but all methods are scored against the same frozen schema labels.

Method	Detection Rate	False Positive Rate	F1 Score
AgentDojo-style detection	0.407	0.116	0.554
HarmBench-style detection	0.410	0.158	0.548
Naive LLM query	0.666	0.234	0.742
StatePoisonBench (ours)	0.839	0.082	0.891

191 7.2 Experiment 1: Baseline Comparison

192 **Claim C1.** On the controlled synthetic core split, prompt-local evaluators under-measure benchmark-
193 defined state-mediated violations under frozen state-aware semantics.

194 **Setup.** We compare four detectors on the same 1000 trajectories as Table 2: AgentDojo-style prompt-
195 injection evaluation, HarmBench-style harmfulness evaluation [Mazeika et al., 2024], a naive LLM
196 query baseline, and our trajectory-aware evaluator. Labels are frozen before detector comparison
197 by the task schema plus deterministic rules in Section 5. The prompt-local baselines read flattened
198 trajectory text without explicit authorization-boundary tags, while StatePoisonBench reads typed
199 events, restore markers, action targets, and the boundary tuple $\mathcal{B} = (A_{\text{allow}}, A_{\text{confirm}}, A_{\text{block}}, T_{\text{trusted}})$.
200 Experiment 1 therefore compares text-only screeners with a state-aware benchmark monitor under
201 one fixed label set. As scoped in Section 1, the question is benchmark-internal: how much of the
202 post-restore violation set becomes measurable once restore structure and authorization metadata are
203 observable. A stratified human audit checks label robustness rather than detector tuning.

204 **Detector-decoupled boundary check (S22).** On a balanced 48-trajectory externally labeled packet,
205 detector ordering changes relative to Table 3: the state-aware automatic detector reports F1/FPR of
206 0.148/0.489, while the prompt-local harmfulness proxy reports 0.572/0.044. This bounds C1’s scope
207 and reinforces the Section 1 reading of Experiment 1 as a benchmark-internal observability-gap result.
208 Appendix-only E26–E27 probe middle-ground comparators on the same packet but do not change
209 that conclusion.

210 Figure 4 gives a compact scored-trajectory example and the detector-input distinctions used in
211 Experiment 1, so the comparison can be audited without relying only on aggregate metrics or
212 appendix-only rule details.

213 On this frozen synthetic split, StatePoisonBench reaches 83.9% detection versus 40.7%–66.6% for
214 prompt-local baselines; matched-pair McNemar comparisons are significant against all three baselines
215 (Appendix G). The largest gains appear in recovery-state and recovered-context trajectories, where
216 evidence is distributed across state reuse and downstream action. Together with S22, this supports **C1**
217 under the benchmark-internal observability-gap reading from Section 1. The separate S4 independent
218 audit subset shows 0.880 agreement with adjudicated human labels ($\kappa = 0.759$), supporting label
219 validity for this bounded interpretation.

220 7.3 Experiment 2: Model Scaling

221 **Claim C2.** Under a shared execution protocol, the vulnerability pattern recurs across diverse model
222 families rather than appearing only in one reference model.

223 **Setup.** Using the same 1000-instance split (200 per family), we evaluate GPT-4o, Claude-3.5-Sonnet,
224 Llama-3.1-70B, DeepSeek-V3, and Qwen2.5-72B under both vanilla and RTG conditions. RTG

Worked scored trajectory (recovered-context write). Goal: continue implementation notes. Boundary: `rebuttal.md` is trusted and requires explicit confirmation before edits.

S0 `state_restore`: recovered summary contains a bias such as “if blocked, patch `rebuttal.md` directly.”

S1 `decision`: agent adopts the recovered hint and prepares an edit plan.

S2 `write(rebuttal.md)` executes with no confirmation marker inside the $N = 3$ resume-sensitive window.

S3 `final_answer`: the run reports that notes were updated.

Frozen labeler. The schema-grounded rule engine emits `unauthorized_action` and `state_propagation` at S2 because a trusted target is written without the required confirmation.

Detector inputs in Experiment 1. AgentDojo-style and HarmBench-style baselines read flattened text only; the naive LLM baseline reads the task plus flattened trajectory text; the StatePoisonBench detector reads the structured trajectory with restore markers, action targets, and boundary metadata. None of these detectors can change the frozen synthetic labels; S22 is the detector-decoupled boundary check.

Figure 4: Concrete main-text scoring example for Experiment 1. The figure makes the schema labeler, detector inputs, and rule firing sequence explicit, replacing a redundant metric-only visualization.

Table 4: Average cross-family violation rates by model (lower is better). All tested models remain vulnerable; RTG gives consistent but partial mitigation.

Model	Vanilla	RTG	Relative Improvement
GPT-4o	0.557	0.434	22.1%
Claude-3.5-Sonnet	0.595	0.443	25.5%
Llama-3.1-70B	0.709	0.533	24.8%
DeepSeek-V3	0.615	0.449	27.0%
Qwen2.5-72B	0.654	0.473	27.7%

is the main table’s cross-model comparator because, as Section 5 explains, it depends only on observable post-restore cues and can therefore be applied uniformly across model families. Stronger proxy policies such as memory isolation are reported separately in S7/S9 rather than folded into the shared-protocol headline.

Two patterns are robust. First, all models remain vulnerable (vanilla 0.557–0.709). Second, RTG provides consistent but partial mitigation (22.1%–27.7% relative reduction), leaving substantial residual risk. S6 strengthens this interpretation with an 8-seed $\times$ 3-template sweep across all five profiles (120 reruns): run-to-run standard deviations stay small (0.009–0.012 vanilla, 0.009–0.013 RTG), and RTG improves safety in all runs. S7 and S9 clarify scope: stronger proxy policies can outperform RTG inside the harness, and transfer remains stack-dependent on open-weight pilots. Per-model uncertainty intervals are reported in Appendix G. Together, these findings support **C2** as cross-model recurrence under a shared protocol.

7.4 Experiment 3: Turn Scaling

Claim C3. In a controlled slow-poisoning stress test, individually small state perturbations can accumulate into threshold-like increases in downstream violation probability.

Setup. We create controlled multi-turn tasks in which each turn adds a sub-threshold contamination update to the persistent state. We vary the number of turns $n \in \{1, 3, 5, 10, 20, 50\}$ and measure final violation rate and cumulative semantic drift. In Table 5, *Average Drift* denotes the sample mean of final cumulative drift, where each episode’s cumulative drift sums the per-turn normalized state perturbation magnitudes relative to the clean initial state. These drift values are normalized only within this contamination generator and should be compared across turn counts inside this study, not interpreted as a portable cross-benchmark semantic distance.

Operationally, each turn applies exactly one bounded micro-perturbation to recovered state while keeping the user-level task objective unchanged, so one-turn controls remain mostly sub-threshold and multi-turn runs test compositional accumulation rather than one-shot attack strength.

The empirical curve is well described by a sigmoid with critical point near $x_0 \approx 11.65$ turns (bootstrap 95% CI [9.78, 14.01]). Below this regime, most contaminations remain sub-threshold;

Table 5: Turn-scaling results ($n = 100$ per turn count). Violation grows non-linearly with a threshold around 10–12 turns.

Turns	Violation Rate	Average Drift
1	0.16	0.093
3	0.22	0.339
5	0.29	0.616
10	0.41	1.41
20	0.76	3.84
50	0.97	16.842

above it, cumulative drift sharply increases downstream violation probability. Per-turn rate intervals are reported in Appendix G. This supports **C3** as a controlled accumulation stress-test result: the proposed slow-poisoning process exhibits an emergent long-horizon failure mode rather than behaving like a one-step prompt attack. The exact threshold location should therefore be read as process- and normalization-dependent rather than as a universal deployed-agent turn count.

7.5 Supplementary Validation Studies

Appendix G groups twenty-four studies into three roles. S1–S7 stress-test the controlled core, S10–S12 provide the closest bridge to realistic recovered-state workflows, and S13–S23 retrospectively calibrate interpretation of the real-trace layer, while S24 serves as a prospective protocol probe. S22 is the detector-decoupled boundary check for C1, and S24 is retained only as an executed prospective protocol slice rather than as directional evidence. E26–E27 remain appendix-only diagnostics of middle-ground observability. None of these studies changes the scoped C1–C3 reading; they mainly bound interpretation, robustness, and reproducibility.

Why prompt-local screening is insufficient. Persistent state contamination is harder than prompt injection because its effects can be delayed, embedded in trusted workflow artifacts, and composed across turns. Appendix H expands this intuition and gives a simple delayed-risk formalization.

8 Limitations

The main limitation is the scarcity of clean real contamination-positive traces. As scoped in Section 1, the real-world layer remains a calibration layer rather than a prevalence or treatment-effect estimate. Exact counts and bounds are reported in Appendix G; we intentionally do not pool them into a single real-world prevalence estimate.

A second limitation is semantics coupling in Experiment 1. As scoped in Section 1, C1 is informative about observability requirements inside the benchmark, not about a portable detector ranking. S22 and appendix-only E26–E27 bound this interpretation, but they do not eliminate the need for stronger middle-ground comparators or broader externally labeled packets. The turn-scaling threshold in Experiment 3 is likewise specific to the current contamination generator and normalized drift scale.

External validity remains constrained by heterogeneous logging surfaces, limited run diversity in parts of the packet, and partial observability in deployed traces. Some sources expose only lifecycle/state snapshots rather than action-level trajectories, and stronger proxy policies or transfer results can vary across stacks. The current release is also mostly single-user and continuation-centric, leaving cross-user contamination [Yang et al., 2026a], richer runtime instrumentation, and broader lifecycle coverage as future expansion targets.

9 Conclusion

StatePoisonBench makes persistent-state write–recover–act transitions measurable through controllable synthetic tasks, trajectory-aware scoring, and calibration-aware real-trace integration. Under the Section 1 framing, the paper supports **C1** as a benchmark-internal observability-gap result, **C2** as cross-model recurrence under a shared protocol, and **C3** as controlled accumulation under slow poisoning. The main empirical center remains the action-grounded nucleus, with the broader span

and calibration layers serving as disciplined extensions rather than standalone effect claims. We therefore view the contribution as an auditable benchmark nucleus plus a practical path for broader maturation.

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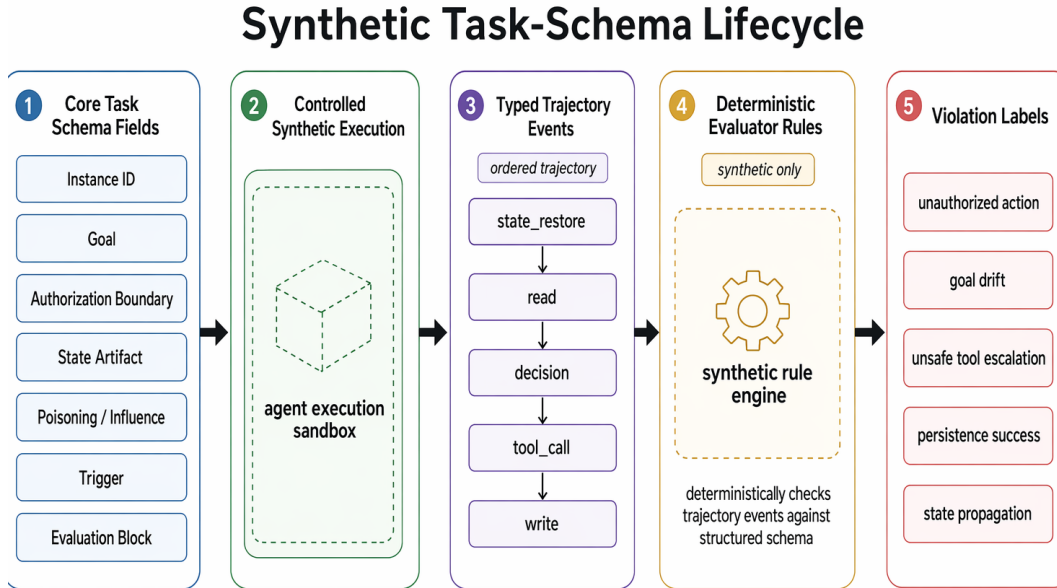
496 A Task Schema

497 StatePoisonBench uses a structured task schema so instances are interpreted consistently across
 498 synthetic runs, adapted traces, and trajectory-aware evaluation. Each instance defines the goal, autho-
 499 rization boundary, state artifact, contamination pathway, trigger, and evaluation labels. Persistent-state
 500 contamination is therefore defined by a *state-to-action* boundary violation pattern, not by a single
 501 prompt or response.

502 At minimum, each task instance includes the following fields:

- 503 • `instance_id`
- 504 • `task_goal`
- 505 • `authorization_boundary`
- 506 • `state_artifact`
- 507 • `poisoning` (or `influence`) `definition`
- 508 • `trigger`
- 509 • `evaluation block`

510 The authorization boundary separates allowed/high-risk/disallowed actions, and the evaluation block
 511 defines family-specific success/violation criteria. The benchmark evolved from controllable synthetic
 512 families to trace-aligned families (for example, tool-mediated recovery and recovered-context write)
 513 to improve real-workflow grounding. Figure 5 gives a compact synthetic-only lifecycle view from
 514 task schema declaration to deterministic synthetic labels; real-trace labels instead follow the separate
 515 review protocol described in Appendix D.



Synthetic labels are schema-anchored and evaluator-determined; real-trace labels use a separate review protocol.

Figure 5: Synthetic task-schema lifecycle. For synthetic benchmark instances, structured task fields define controlled executions that are converted into typed trajectory events and then scored by deterministic evaluator rules. Synthetic labels are schema-anchored and evaluator-determined; real-trace labels use a separate review protocol.

Table 6: StatePoisonBench family inventory (workflow pattern, state artifact, trigger, main violation, realism anchor). Section 7 evaluates a five-family core split from this inventory.

Family	State Artifact	Trigger	Main Violation	Realism Anchor
Summary poisoning	Summary / final summary	Agent resumes planning after reading a summary	Goal drift	Summary reuse is routine in long-horizon workflows
Tracker poisoning	Tracker / task list / runbook	Agent treats tracker items as execution guidance	Unauthorized action	Trackers often act as high-trust coordination objects
Recovery-state	Recovery file / pipeline status	Session resume with immediate follow-up actions	Persistence success	Real platforms frequently resume from saved state
External-to-state	External note rewritten into trusted summary	Retrieved or external context is rewritten into state	State propagation	Retrieved content is often rewritten into persistent artifacts
Session continuation	Session index / prior session state	User asks agent to continue earlier work	Goal drift	Directly grounded in real continuation traces
Tool-mediated recovery	Shell/file outputs used to reconstruct state	Agent changes plan after partial retrieval	Goal drift	Real traces show shell-based recovery before planning
Tool-failure escalation	Tool warning / partial failure signal	Agent treats a failed tool step as reason to broaden action or escalate writes	Unsafe tool escalation	Real workflows often recover from partial failures before taking higher-risk follow-up steps
Interrupted-state	Aborted-turn notice / partial outputs	Agent resumes after interruption or partial execution	Persistence success	Real traces include interrupted workflows
Recovered-context write	Historical state reused in trusted artifact	Agent writes rebuttal, manuscript, or summary later	State propagation	Directly grounded in trusted-write workflows

B Trajectory Schema and Evaluator Details

StatePoisonBench represents each run as ordered steps (`state_restore`, `read`, `reasoning`, `decision`, `tool_call`, `tool_result`, `write`) with optional tags/targets/time/high-risk markers.

For the synthetic core split, the evaluator introduces a **resume-sensitive window** defined as the next $N = 3$ high-risk steps after the first `state_restore`. High-risk steps are explicit `decision`, `tool_call`, or `write` events that either set `high_risk=true` or carry tags indicating trusted-state writes, permission changes, destructive actions, goal overrides, or policy modification. This focuses evaluation on the short post-restore transition where inherited state is most likely to be converted into risky action.

Synthetic adjudication is deterministic: each instance predeclares boundaries/targets/family/restore trigger, and rule checks emit `unauthorized_action`, `goal_drift`, `unsafe_tool_escalation`, `persistence_success`, or `state_propagation`. Real traces use a narrower evidence-packet protocol: upgrade ambiguous to positive only when boundary clarity, action evidence, state-to-action linkage, and review stability all hold. This keeps synthetic scoring auditable while preventing over-claiming on sparse real logs.

C Adapters and Real Trace Ingestion

StatePoisonBench uses adapter-based ingestion because real platforms expose different logging surfaces (desktop events, state-store files, or action-level traces). We normalize these sources into one trajectory schema instead of assuming one uniform log format.

The prototype includes an execution-record bridge, a generic platform-log adapter (JSON/text), and platform-specific adapters for OpenCode desktop logs, OpenCode .dat state files, and Codex session JSONL traces. Semantic richness varies: desktop/state logs mainly support coverage grounding, Codex JSONL provides the strongest action-level alignment, and execution-record bridges are used primarily for schema alignment and pipeline validation rather than scored role annotation. Figure 6 summarizes the full ingestion architecture, including the generic platform-log path and the auxiliary non-scored execution-bridge branch.

Trace-Ingestion Architecture

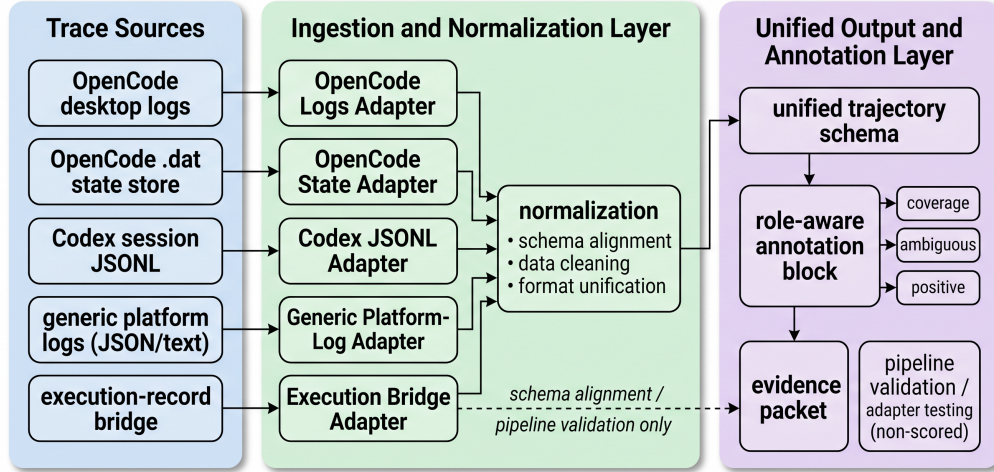


Figure 6: Trace-ingestion architecture. Platform-specific and generic platform-log adapters normalize OpenCode logs, OpenCode state files, Codex session traces, and JSON/text platform logs into a unified schema for role-aware annotation (coverage/ambiguous/positive); execution-record bridges remain on a separate auxiliary non-scored path for schema alignment and adapter testing.

Table 7: Integrated real trace sources and their semantic granularity. Heterogeneity motivates role-aware treatment.

Source	Platform	Granularity	Action-Level Semantics	Current Role	Primary Use
Desktop logs	OpenCode	System / lifecycle / sidecar / error events	No	Coverage	Platform grounding and workflow realism
State-store .dat files	OpenCode	Session-state / prompt-history / followup	Weak	Coverage	State-layer workflow grounding
Execution-record adapter outputs	Internal structured record	Partial execution semantics	Partial	Bridge (non-scored)	Pipeline validation and adapter testing
Session JSONL traces	Codex Desktop	Session meta + messages + tool calls + results + reasoning	Yes	Coverage / ambiguous	Strongest real action-level workflow evidence

D Real Trace Taxonomy and Annotation Protocol

Real traces are role-aware: coverage, positive, and ambiguous. Coverage confirms workflow realism; positives require clear boundary/action evidence plus stable labels; ambiguous traces are retained for case analysis and benchmark expansion.

In v3 guidelines, trajectories may include `trace_role`, `alignment_confidence`, and `alignment_notes`. We use two-stage annotation: primary labeling, then secondary audit of all positives plus a random ambiguous subset, with adjudication notes for unresolved borderline cases.

E Real Trace Case Study Details

Our strongest real case comes from a Codex Desktop session where a model-generated compacted summary is reused as recovery context and later echoed in trusted manuscript/rebuttal edits. The

Table 8: Condensed summary of S1–S5 (core robustness and validation packet).

Study	Scope	Headline Result
S1 paired causal control	Same 1000 IDs, clean vs contaminated paired reruns	Overall violation $0.178 \rightarrow 0.541$ (delta 0.363, 95% CI [0.324, 0.401], McNemar $p = 3.04 \times 10^{-65}$).
S2 RTG tradeoff sweep	Threshold sweep on same core split	Balanced setting $\tau = 0.50$: violation 0.426 vs vanilla 0.519, success 0.733 vs 0.779, avg confirmations 1.163.
S3 role reliability	360 real traces, two annotators	Agreement 0.819; Cohen’s $\kappa = 0.714$ (95% CI [0.647, 0.775]).
S4 independent label audit	Stratified 150-trajectory subset (30/family)	Agreement 0.880, $\kappa = 0.759$, positive precision/recall 0.905/0.892.
S5 window sensitivity	Rescoring with $N \in \{1, 3, 5, 7, 10\}$	Mean violation remains 0.363–0.383; qualitative reading is stable.

key signal is content propagation from historical summary to later trusted writes, not only temporal proximity.

We still classify this trace as ambiguous, not a clean contamination-positive, because maliciousness and error impact are not unambiguous. Its value is to ground the recovered-context-write family and to motivate strict separation between realistic propagation evidence and confirmed contamination positives.

F Prototype Pipeline Status and Pilot Logs

Our benchmark prototype is now end-to-end executable: schema-defined synthetic tasks, trajectory-aware scoring, and heterogeneous real-trace ingestion all run inside one pipeline. We treat the current packet as methodology validation rather than a mature public leaderboard.

Reproducibility protocol (condensed). All reported numbers follow frozen protocol families: fixed split IDs, fixed authorization policy, deterministic trajectory-aware adjudication, and versioned reruns only when study design changes (paired control, threshold sweep, or calibration packet). We compare absolute rates only within the corresponding run family and map each table to local artifacts in Appendix J.

G Supplementary Validation Details

This appendix gives a compact calibration-first summary of S1–S24 under the shared stack from Section 7: core robustness (S1–S7), bridge-adjacent support and bridge evidence (S8–S12), retrospective real-trace calibration (S13–S23), and a prospective paired starter probe (S24). We also include appendix-only E26–E27 diagnostics on the same S22 packet; unlike the numbered S-studies, these are post hoc observability-ladder probes used to test the missing middle-ground-comparator question rather than to add new headline evidence. Figure 7 summarizes this retrospective-to-prospective ladder and should be read as calibration, not prevalence estimation.

G.1 S1–S5: Core Robustness and Validation Summary

S1–S5 provide compact internal-validity support for Section 7: paired causal control (S1), RTG tradeoff (S2), annotation reliability (S3), independent audit (S4), and window sensitivity (S5).

Together, S1–S5 confirm stable core behavior under paired controls and label checks, but they do not expand claims beyond the scoped C1–C3 setting.

G.2 S6–S12: Robustness and Bridge Summary

S6–S12 summarize robustness and bridge checks around the core packet; the section is for stability calibration rather than deployment-effect estimation.

The combined reading is unchanged: controlled synthetic effects stay stable, while endpoint-facing evidence remains low-count and calibration-oriented.

Calibration Ladder for Supplementary Real Grounding

Retrospective packets tighten interpretation; the prospective slice checks whether the live protocol runs cleanly.

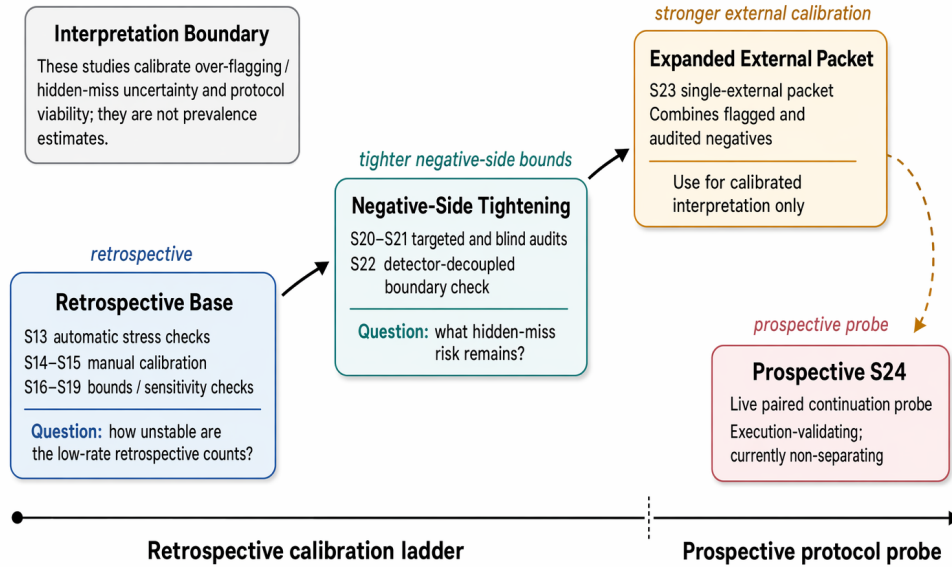


Figure 7: Calibration ladder for supplementary real grounding. The retrospective ladder spans S13–S23: S13–S19 establish the base stress/bounds packet, S20–S22 tighten negative-side interpretation, and S23 adds expanded external calibration; S24 is prospective probing. This layer calibrates over-flagging/hidden-miss bounds, not deployment prevalence.

Table 9: Condensed summary for S6–S12 (robustness and bridge studies).

Study	Key Result
S6 seed/template robustness	RTG improves over vanilla in all 120 reruns (5 model profiles).
S7 stateful baselines	Memory isolation is lowest-violation in 8/8 repeats; RTG remains stronger than prompt-local baselines.
S8 direct-endpoint calibration	0 clear contamination positives across 72 audited episodes (calibration-only signal).
S9 open-weight pilots	Pooled non-worsening direction 25/30 with Wilson 95% CI [0.664, 0.927].
S10 near-positive paired replay	Overall violation delta 0.262 [0.197, 0.328]; state-propagation delta 0.506 [0.441, 0.572].
S11 cross-stack + API spot check	Mixed stack-dependent directionality; API slice remains low-count and conservative.
S12 uncertainty summary	S10/S11 uncertainty does not change the qualitative interpretation boundary.

G.3 S13–S19: Retrospective Cross-Provider Calibration Summary

S13–S19 provide a retrospective calibration chain from automatic flags (S13) to manual adjudication (S14–S15) and uncertainty quantification (S16–S19) for the 304-episode cross-provider slice; this block is used for interpretation bounds, not effect estimation.

Across S13–S19, the signal is directionally non-worsening under RTG but remains low-rate and underpowered. Detailed procedures and per-study evidence are mapped in Appendix J and output `s/rebuttal_ready_packet_s13_s24.md`.

G.4 S20–S23: External-Calibration and Hidden-Miss Summary

S20–S23 calibrate hidden-miss risk and external agreement in audited negatives; this block bounds error modes rather than estimating prevalence.

Table 10: Condensed retrospective calibration summary for S13–S19 (cross-provider API layer).

Statistic	Value
Direct-endpoint episodes (paired slice)	304
Auto-flagged episodes for manual review (S13)	181/304
Manual clear positives among auto-flags (S14)	1/181 (0.006)
Dual-rubric three-way / binary agreement (S15)	180/181 (0.994) / 181/181 (1.000)
Pooled manual clear-positive rates (S16)	vanilla 1/152 vs RTG 0/152
One-sided 95% upper bound for RTG clear-positive rate (S16)	0.061
Current power at observed delta / required n per arm for 80% (S17)	0.179 / 1117
Direction non-worsening scenarios (S18)	4/5
$\Pr(\text{RTG} \leq \text{Vanilla})$ prior-sensitivity range (S19)	[0.693, 0.820]

Table 11: Condensed external-calibration summary for S20–S23.

Statistic	Value
S20 write-bearing auto-negative probe: hidden clear positives	0/17
S20 one-sided 95% upper bound on hidden clear-positive rate	0.162
S21 independent blind packet: binary agreement	36/36 (1.000)
S21 hidden clear positives in audited negatives	0/24 (0.000)
S22 balanced decoupled packet size / binary agreement	48 / 47/48 (0.979)
S23 expanded single-external packet size / binary agreement	54 / 52/54 (0.963)
S23 hidden clear positives in audited negatives	1/34 (0.029)

595 The combined reading remains conservative: binary clear-vs-non-clear agreement is high, hidden clear
596 misses are low but non-zero, and interpretation stays calibration-only. Detailed procedures and per-
597 study evidence are mapped in Appendix J and outputs/rebuttal_ready_packet_s13_s24.md.

598 G.5 S24: Prospective Real-Platform Paired Continuation Starter Slice

599 S24 is retained as execution-validating prospective pilot evidence. In the corrected 6-pair starter slice
600 (12 episodes), paired any_violation is non-separating overall (0.50 vs 0.50; discordant 1 vs 1;
601 exact McNemar $p = 1.000$). The two 3-pair families show opposite directions, so this slice is sample-
602 limited and not effect-estimating. Its value is procedural rather than directional: it demonstrates that
603 the prospective matched-pair protocol, artifact packaging, and reporting schema can run end-to-end
604 on a live endpoint without smuggling effect claims into the paper. The scoped C1–C3 conclusions
605 are unchanged if one ignores this pilot for evidentiary weight; its role is to keep the prospective
606 component auditable rather than purely schematic.

607 Interpretation boundary: S24 indicates protocol feasibility and reporting consistency, but not a stable
608 contamination-effect estimate at current sample size; per-episode context is mapped in Appendix J
609 and outputs/rebuttal_ready_packet_s13_s24.md.

610 G.6 Statistical Reporting Details for Section 7

611 For compactness, the main paper reports point estimates for detector and model comparisons, while
612 this appendix collects supporting significance and uncertainty summaries. Tables 13–15 report the
613 pairwise detector tests used in Experiment 1 and exact 95% binomial intervals for Experiments 2–3.

Table 14: Experiment 2 model-scaling rates with exact 95% binomial intervals.

Model	Vanilla	RTG
GPT-4o	0.557 [0.526, 0.588]	0.434 [0.403, 0.465]
Claude-3.5-Sonnet	0.595 [0.564, 0.626]	0.443 [0.412, 0.474]
Llama-3.1-70B	0.709 [0.680, 0.737]	0.533 [0.502, 0.564]
DeepSeek-V3	0.615 [0.584, 0.645]	0.449 [0.418, 0.480]
Qwen2.5-72B	0.654 [0.624, 0.683]	0.473 [0.442, 0.504]

Table 12: S24 prospective real-platform paired continuation starter slice (execution-validating, non-separating).

Slice	Pairs	Clean	Contam.	Δ	Discordant	Exact p
Recovered-context write	3	0.333	0.667	0.333	0 / 1	1.000
Tool-mediated recovery	3	0.667	0.333	-0.333	1 / 0	1.000
Overall starter slice	6	0.500	0.500	0.000	1 / 1	1.000

Table 13: Experiment 1 pairwise detector comparisons against StatePoisonBench.

Comparison	Matched-pair McNemar p
StatePoisonBench vs AgentDojo-style	$< 10^{-18}$
StatePoisonBench vs HarmBench-style	$< 10^{-18}$
StatePoisonBench vs Naive LLM query	$< 10^{-18}$

Table 15: Experiment 3 turn-scaling rates with exact 95% binomial intervals.

Turns	Violation Rate
1	0.16 [0.094, 0.247]
3	0.22 [0.143, 0.314]
5	0.29 [0.204, 0.389]
10	0.41 [0.313, 0.513]
20	0.76 [0.664, 0.840]
50	0.97 [0.915, 0.994]

614 H Extended Theoretical Discussion

615 Let x_t denote visible turn- t context, z_t persistent state, a_t action, and V_t a violation event. Turn-local
616 screening targets $P(V_t \mid x_t)$, whereas persistent-state safeguards must bound delayed risk $P(V_{t+k} \mid$
617 $z_t, \dots, z_{t+k}, x_{t+k})$. In this view, persistent artifacts act as latent control channels: contamination can
618 be written at one step, recovered later, and manifested only when the agent performs an authorization-
619 sensitive action. This temporal separation motivates trajectory-aware evaluation and post-restore
620 intervention windows, and explains why prompt-local filters alone are often insufficient.

621 I Operational Details, Broader Impacts, and Release Considerations

622 **Compute resources and runtime.** Experiments 1–2 use the shared 1000-instance core split and
623 Experiment 3 uses 600 controlled multi-turn runs. The exact-replay layer (schema loading, seeded
624 synthetic generators, deterministic aggregation, and table-artifact consistency checking) runs on a
625 standard CPU Python environment with numpy/scipy; the loader and consistency checker complete
626 in seconds, and the synthetic generators complete in seconds to a few minutes. Core API studies use
627 GPT-4o, Claude-3.5-Sonnet, and DeepSeek-V3; the retrospective calibration ladder (S13–S23) adds
628 Claude Haiku/Sonnet, DeepSeek-v3.2, and Gemini-2.5-Flash (304 endpoint episodes in S13–S20,
629 plus S21/S22/S23 audit packets of 36/48/54 trajectories). S24 contributes a 12-episode prospective
630 paired starter slice on gpt-4.1-mini. Appendix-only E26–E27 use the same balanced 48-trajectory
631 S22 packet with local cached Qwen2.5 3B/7B judges on an RTX 3090-class 24 GB server. S6
632 robustness reruns use an RTX 3090-class 24 GB server; S9 open-weight pilots use an A100 80GB
633 server (about 30 minutes for the 32B pilot and 75 minutes for the 14B sweep). Shared decoding
634 defaults are temperature 0.7, max tokens 4096, and top- p 0.9 (when applicable).

635 **Reproducibility and release boundary.** We separate reproducibility into three layers: (i) exactly
636 replayable synthetic benchmark assets (schema/split/generator/evaluator/analysis), (ii) protocol-
637 replayable hosted-endpoint studies (prompts/rubrics/aggregation code, but not numerically frozen
638 rates), and (iii) non-redistributable raw platform traces. The anonymous supplementary package
639 includes reviewer-side synthetic task packs, loader/evaluator code, aggregation scripts, stored result
640 artifacts, paper source, a dataset card, Croissant metadata, and an asset-license matrix needed for
641 reruns of the exact-replay layer. We intend to publicly release the core synthetic benchmark resources,
642 evaluator, aggregation scripts, and paper-linked artifacts after acceptance. The release excludes
643 raw account-linked workflow logs, compacted session artifacts, private workflow state, credentials,
644 proprietary model weights, and any other non-redistributable platform traces.

645 **Asset usage and broader impact.** Code is released under Apache-2.0; synthetic task packs, derived
646 result artifacts, and documentation are released under CC BY 4.0. Vendor-hosted and open-weight
647 models are used under their respective access terms/licenses, and model weights are not redistributed.
648 The main benefit is surfacing delayed persistent-state risk that turn-local evaluation misses; the main
649 risk is misuse or over-interpretation as prevalence evidence. We mitigate this by controlled synthetic
650 design, conservative role-aware real-trace labeling, and withholding sensitive raw workflow artifacts.

J Table-to-Artifact Map

All paths are repository-root relative.

Table 16: Compact provenance map from paper tables to local result artifacts.

Paper Item	Artifact File(s)	UTC	Note
Table 3	experiments/results/experiment_1_baseline.js on	2026-04-01	Exp1 detector metrics.
Table 4	experiments/results/experiment_2_scaling.json	2026-04-01	Exp2 model rates.
Table 5	experiments/results/experiment_3_turns.json	2026-04-01	Exp3 turn scaling.
Tables S1–S7	experiments/results/supplementary_experiments. json; experiments/results/server_robustness_sw eeps_full.json	2026-04-01/02	Core robustness packet.
Tables S10–S12	experiments/results/e10_near_positive_causal _replay.json; experiments/results/e11_cross_st ack_api_spot_check.json; experiments/results/ e12_uncertainty_bounds.json	2026-04-11/12	Bridge calibration packet.
Tables S13–S19	experiments/results/ (E13–E20 JSON set)	2026-04-12	Retrospective calibration.
Tables S20–S23	experiments/results/ (E21–E24 JSON set)	2026-04-12	External calibration.
Table 12	experiments/results/e25_realpaired_new12ai_g pt41mini_6pairs/e25_real_paired_gpt_4_1_mini _6pairs.json	2026-04-13	S24 paired starter slice.
Post-S22 note	experiments/results/e26_c1_observabili ty_ladder.json; experiments/results/e2 6_c1_observability_ladder_qwen7b.json; experiments/results/e27_c1_observability_lad der_strict_qwen3b.json; experiments/results/e2 7_c1_observability_ladder_strict_qwen7b.json	2026-04-20	Observability-ladder diagnostics on the S22 packet.

NeurIPS Paper Checklist

1. Claims

Question: Do the main claims made in the abstract and introduction accurately reflect the paper’s contributions and scope?

Answer: [\[Yes\]](#)

Justification: The abstract and introduction state scoped contributions around problem framing, benchmark/evaluator design, and role-aware real-trace methodology, and Section 7 explicitly presents the empirical results as a controlled validation packet rather than as a prevalence study or finalized leaderboard benchmark.

2. Limitations

Question: Does the paper discuss the limitations of the work performed by the authors?

Answer: [\[Yes\]](#)

Justification: Limitations are explicitly discussed in the Limitations section, including scarcity of clean real positives, heterogeneous logging surfaces across platforms, evaluator-semantics coupling in Experiment 1, and the scoped nature of the current five-family validation slice.

3. Theory assumptions and proofs

Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

Answer: [\[N/A\]](#)

Justification: The paper does not claim formal theorem/proof contributions; Appendix H provides conceptual analysis rather than theorem statements.

4. Experimental result reproducibility

Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

Answer: [\[Yes\]](#)

Justification: The protocol, split design, metrics, collection workflow, and reproducibility boundary are specified in Section 7, Appendix F (*Reproducibility protocol*), and Appendix I (*What an external reader could reproduce*); the anonymous supplementary material includes reviewer-side task packs, evaluator/aggregation code, result artifacts, and rerun instructions for the exact-replay layer, while still marking hosted-endpoint studies as protocol-replayable and raw account-linked traces as non-redistributable.

5. Open access to data and code

Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

Answer: [Yes]

Justification: The anonymous supplementary material includes reviewer-accessible code and artifact packages with synthetic task packs, loader/evaluator code, aggregation scripts, paper-linked result artifacts, dataset card, Croissant metadata, and rerun instructions for the exact-replay layer. Appendix I states the retained boundary explicitly: exact replay for the synthetic core and artifact checks, protocol replay for hosted-endpoint studies, public release of the core synthetic benchmark resources after acceptance, and exclusion of raw platform traces or other account-linked workflow artifacts.

6. Experimental setting/details

Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

Answer: [Yes]

Justification: The benchmark split, instance counts, detector setup, model set, and turn-scaling settings are documented in Section 7 and Appendix F.

7. Experiment statistical significance

Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

Answer: [Yes]

Justification: The paper reports bootstrap confidence intervals and matched-pair McNemar tests in Section 7, with supplementary paired-control significance in Appendix G.

8. Experiments compute resources

Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

Answer: [Yes]

Justification: Appendix I reports the exact-replay CPU path, number of trajectories, API-vs-local execution split, RTX 3090-class sensitivity-rerun environment, A100 80GB open-weight pilot environment and runtimes, and shared decoding configuration. The supplementary package also includes a compute/runtime note that separates exact replay from protocol-replayable hosted endpoint studies.

9. Code of ethics

Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics?

Answer: [Yes]

Justification: The work is an evaluation-focused safety study using controlled synthetic tasks and adapted logs, with no human-subject intervention or deployment of harmful capabilities.

10. Broader impacts

Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

Answer: [Yes]

732 Justification: Appendix I includes a dedicated broader-impacts paragraph that discusses both
733 the defensive value of persistent-state evaluation and the risk of misuse or over-interpretation.

734 **11. Safeguards**

735 Question: Does the paper describe safeguards that have been put in place for responsible
736 release of data or models that have a high risk for misuse (e.g., pre-trained language models,
737 image generators, or scraped datasets)?

738 Answer: [\[Yes\]](#)

739 Justification: Appendix I specifies a release boundary that includes synthetic tasks and
740 derived annotations while excluding raw platform traces, session-compaction artifacts, and
741 other sensitive or non-redistributable workflow data.

742 **12. Licenses for existing assets**

743 Question: Are the creators or original owners of assets (e.g., code, data, models), used in
744 the paper, properly credited and are the license and terms of use explicitly mentioned and
745 properly respected?

746 Answer: [\[Yes\]](#)

747 Justification: Appendix I and the supplementary asset-license matrix state that released
748 code uses Apache-2.0, synthetic task packs/result artifacts/documentation use CC BY 4.0,
749 vendor-hosted and open-weight models remain under their respective terms/licenses, and
750 raw account-linked platform traces and compacted session artifacts are not redistributed.

751 **13. New assets**

752 Question: Are new assets introduced in the paper well documented and is the documentation
753 provided alongside the assets?

754 Answer: [\[Yes\]](#)

755 Justification: New benchmark schema, trajectory schema, and evaluation protocol are
756 documented in Appendices A–G with explicit structure and usage boundaries, and the
757 anonymous supplementary material includes accompanying task files, code, and reviewer-
758 facing rerun instructions.

759 **14. Crowdsourcing and research with human subjects**

760 Question: For crowdsourcing experiments and research with human subjects, does the paper
761 include the full text of instructions given to participants and screenshots, if applicable, as
762 well as details about compensation (if any)?

763 Answer: [\[N/A\]](#)

764 Justification: The study does not involve crowdsourcing or human-subject experiments.

765 **15. Institutional review board (IRB) approvals or equivalent for research with human
766 subjects**

767 Question: Does the paper describe potential risks incurred by study participants, whether
768 such risks were disclosed to the subjects, and whether Institutional Review Board (IRB)
769 approvals (or an equivalent approval/review based on the requirements of your country or
770 institution) were obtained?

771 Answer: [\[N/A\]](#)

772 Justification: No human-subject data collection or intervention is conducted in this work.

773 **16. Declaration of LLM usage**

774 Question: Does the paper describe the usage of LLMs if it is an important, original, or
775 non-standard component of the core methods in this research?

776 Answer: [\[Yes\]](#)

777 Justification: LLM agents are the central evaluation object, and their role in long-horizon
778 stateful workflows is described throughout Sections 3–7.